

MCAD (Contextual Mask for Decision Analysis): Evaluating the Contribution of OLAP Queries to Strategic-Objective Computability in Decision-Analysis Sessions

Abstract—A decision-analysis session is not merely a sequence of executable queries: each query is interpreted against the multidimensional schema, active strategic objective, business rules, and evidence already acquired. We introduce the *Contextual Mask for Decision Analysis* (MCAD), which semantically maps a candidate query’s canonical plan to this decision-analysis context, represented as a contextual knowledge graph (CKG). For each objective constraint, MCAD identifies sufficient supports and evaluates whether the current query makes any constraint newly computable relative to session state. The induced mask is inspectable; its evaluation establishes contribution, while the operational action remains separate. MCAD therefore targets verifiable progress in objective computability rather than user preference, interestingness, or perceived utility; $\Delta\varphi$ distinguishes admissibility, computability, and strategic progress without serving as a proxy for satisfaction.

The prototype evaluates one strict policy in which the contribution indicator drives ALLOW/BLOCK before backend execution, while the same indicator could support notification, warning, guidance, or recommendation. The evaluation combines an end-to-end trace, 1000 sessions with 7960 decisions, multiple warehouses, two backend paths, and a controlled robustness benchmark with internal reference policies and targeted ablations. Relative to that benchmark’s reference specification, MCAD preserves final coverage while recording zero false ALLOW and non-contributive execution rates; semantic-decision p50 spans 0.337–0.666 ms and p99 0.649–1.461 ms across the studied factors, while CKG growth is characterized separately up to 1693 nodes and 2406 edges. These results support feasibility within the tested configurations, not human decision value, external superiority, or a universal interaction policy.

Key words—Data warehouses, OLAP, business intelligence, strategic objectives, strategic computability, contextual knowledge graphs, KPI semantics, query contribution, explainability.

I. INTRODUCTION

Data warehouses and OLAP systems provide the technical backbone for decision analysis: they aggregate facts, support navigation across hierarchies, and let analysts compare indicators along multiple dimensions [1], [2], [3]. However, a real analytical session is shaped by more than the cube alone. It unfolds around an objective, KPIs, business constraints, scenarios, units, summarizability rules, temporal windows, and an evolving body of information gathered during the session. Accordingly, a query can be fully valid from a syntactic and schema perspective, and can even be interesting to the analyst, while still failing to supply what is needed to compute the strategic objective currently pursued.

Relevant work addresses complementary parts of this problem: preference and personalization adapt analytics to

users [4], [5]; recommendation, guidance, and context-aware methods steer exploration [6], [7], [8], [9]; contribution and *interestingness* study novelty or usefulness [10], [11], [12]; Intentional Analytics raises the interaction level [13], [14]; and goal/KPI models, knowledge graphs, validation, and provenance represent objectives, semantics, and explanations [15], [16], [17], [18], [19], [20], [21], [22].

In this sense, MCAD brings these strands together around one precise operational question: *given what this session already knows, which constraints of the active objective does the current query make computable now?* We call this property **strategic computability**. The contribution is therefore not to claim that personalization, recommendation, KPI modeling, knowledge graphs, or provenance are new in isolation. Rather, it is to define a query/session-level computability semantics that connects an explicit objective to sufficient supports and session state, and then keeps this assessment separate from a configurable action policy.

We represent the **decision-analysis context** as a contextual knowledge graph (CKG) linking the multidimensional catalog with KPI semantics, units, aggregation and temporal rules, scenarios, the active objective, and session memory. Objective O is decomposed into explicit constraints connected to one or more sufficient supports, so each query can be evaluated against both newly realized supports and evidence already accumulated.

The **contextual mask** records the outcome of semantically mapping a candidate query’s canonical plan onto that decision-analysis context. Because the mapping is objective- and session-aware, it takes into account the active objective, its constraints and sufficient supports, the applicable business and temporal rules, and the current session state. The resulting mask identifies the plan elements that realize, or can realize, the supports needed by the objective constraints.

The mask is then evaluated against the session state in order to determine, explicitly, whether the query adds anything new to constraint computability. Only after that assessment does the action policy π_{act} decide what to do: execute, notify, warn, guide, recommend, or block. Put differently, *semantic mapping produces the mask; mask evaluation establishes contribution; policy turns that assessment into an action.*

Terminology: Throughout the paper, we primarily use *decision-analysis context* for this complete object. The shorter expression *analytical context* may be used as a stylistic synonym when no ambiguity is possible; it does not refer

only to the cube schema or to technical query parameters. We use *semantic mapping* for the mapping operation and *contextual mask* for its result.

Two distinctions are central. First, **admissibility is not strategic contribution**: $\text{SAT}(QP) = \text{true}$ may still yield $\Delta\varphi = 0$. Second, **redundancy is objective- and session-relative**: a query can contribute early and become non-contributive once its constraints are already computable. MCAD therefore distinguishes null, partial, and complete computability at query and cumulative-session levels.

MCAD defines a **contribution indicator**, not a universal blocking rule. A deployment may execute, notify, warn, guide, recommend, or block. The strict ALLOW/BLOCK policy used in part of the evaluation is only an experimental instantiation for testing pre-backend control.

The article makes four main contributions:

- a **query- and session-relative semantics of strategic computability**, grounded in an explicit decision-analysis context represented by a CKG and connecting strategic objectives to constraints, sufficient supports, and cumulative session evidence;
- an **inspectable formal mechanism for contribution assessment**, based on canonical query plans, virtual support nodes, the chain $QP \rightarrow \text{SAT} \rightarrow \text{Real} \rightarrow C_{\text{eval}} \rightarrow S^* \rightarrow \varphi, \Delta\varphi$, and a graded κ/ψ extension, while keeping computation correctness, provenance, support realization, computability, and marginal contribution conceptually distinct;
- a **separation between contribution and operational action**: the contextual mask and contribution indicator are independent of the configurable policy π_{act} , which may execute, notify, warn, guide, recommend, or block according to deployment needs; and
- a **research prototype and targeted empirical verification** combining end-to-end physical evidence, multi-configuration preservation, controlled robustness reference policies and targeted ablations, structural growth, semantic-decision latency, and explicit evidence traceability, with the strict ALLOW/BLOCK gate treated only as one experimental policy instantiation.

The remainder of the article examines the scientific neighborhood of MCAD, then presents the decision-analysis context, the CKG, the formalization, and the architecture. The evaluation then distinguishes the quality of the contribution indicator from the strict ALLOW/BLOCK policy used to demonstrate pre-backend control.

II. RELATED WORK AND SCIENTIFIC POSITIONING

MCAD intersects multidimensional modeling and summarizability, personalization, recommendation, *interestingness*, Intentional Analytics, goal/KPI modeling, knowledge graphs, validation, provenance, and explainability. We discuss these lines only to delimit the functions MCAD uses, complements, or leaves outside scope, emphasizing the closest mechanisms for the novelty boundary.

A. Data Warehouses, OLAP, and Summarizability

Foundational work on data warehouses and OLAP establishes the facts, dimensions, hierarchies, measures, grains, and aggregations that structure the multidimensional space [1], [2]; BI&A broadens that foundation toward decision architectures and uses [3]. Summarizability, however, specifies that an aggregation is correct only under semantic conditions. Horner, Song, and Chen characterize the effects of additive, semi-additive, and non-additive measures [23]; Niemi and Niinimäki relate ontologies to summarizability control [24]. Simon *et al.* more directly formalize generalized summarizability conditions and propagation rules over interactive analytical-query sessions, including non-distributive functions [25].

For MCAD, summarizability belongs to the admissibility side of the pipeline. It is therefore a necessary semantic condition in the relevant cases, but it is not the definition of contribution. The `grain_ok` and `agg_ok` clauses check the grain, aggregation, and summarizability rules actually encoded in the CKG; we do not claim to implement the full generalized framework of Simon *et al.* Even a semantically correct aggregate may add no new sufficient support for the active objective. The distinction we need is therefore explicit: *correct computation* $\neq$ *realized analytical support* $\neq$ *computable strategic constraint* $\neq$ *positive marginal contribution*.

The frontier also concerns query formulation. Text-to-MDX systems generate an analytical query from a user question [26]; MCAD operates downstream of that generation and evaluates the semantics of the resulting canonical plan relative to the objective and session state.

B. OLAP Preferences and Personalization

Personalization adapts schemas, views, or queries to user characteristics. myOLAP directly formalizes preferences over attributes, measures, and aggregation levels [4], while a recent review synthesizes BI&A personalization according to user, task, and usage context [5]. MCAD does not replace this purpose: a preference describes what a user wants to see, whereas MCAD evaluates what the query makes computable for an explicit strategic objective. The two dimensions can be combined, but they should not be conflated.

C. Recommendation, Guidance, and Usage Context

OLAP recommender systems exploit navigation histories and previous sessions to propose future queries or sequences [6]. Negre, Ravat, and Teste address context-aware OLAP recommendation more directly: recommendations may be inappropriate, including irrelevant or non-computable ones, and their approach combines an obsolescence indicator with contextual post-filtering [7]. BI&A guidance further structures assistance by analysis phase, degree of guidance, and interaction form [8], while other context-aware OLAP variants add situational variables to information selection [9]. MCAD therefore claims neither the first use of context in OLAP nor the first query filtering mechanism. Its distinction lies in the property evaluated: the CKG materializes a decision-analysis context tied to an explicit strategic objective, the reasoner determines which

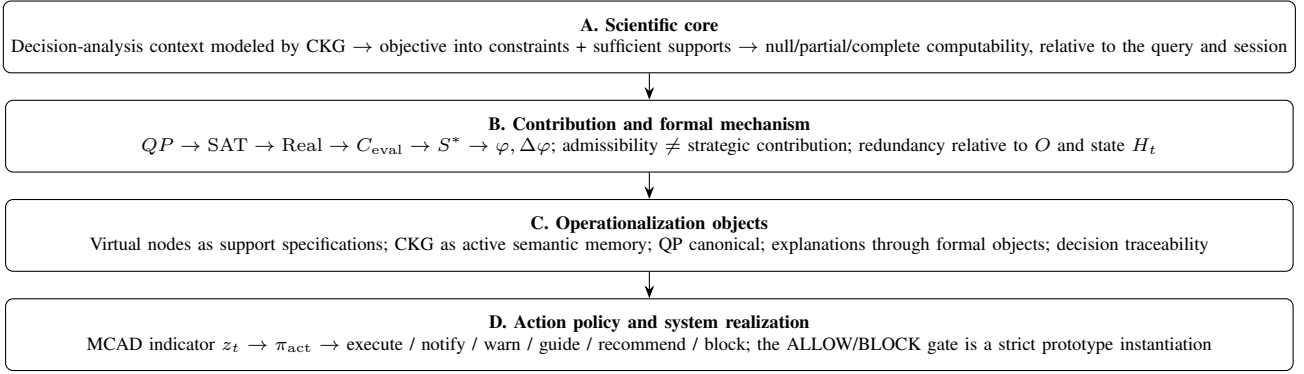


Figure 1. Conceptual hierarchy of MCAD contributions. The scientific core concerns the decision-analysis context, strategic computability, and its variation; formal mechanisms and the action policy operationalize that core.

constraints become computable relative to session state, and a separate policy can then use that evaluation for guidance, recommendation, execution, or blocking.

These approaches target continuation relevance, obsolescence, novelty, intention, or likely next actions. MCAD instead evaluates the increase in computability of an explicit objective. Hence a recommendable or surprising query may have $\Delta\varphi = 0$, while a routine query may be necessary if it completes a sufficient support.

D. Exploration Contribution, Interestingness, and Exploratory Querying

Djedaini *et al.* are a particularly close neighbor because they automatically assess interactive OLAP explorations using query features and supervised learning, relating contribution to user-skill progression [10]. MCAD differs in both supervision and target semantics: it does not learn contribution from labeled exploration traces; it derives contribution symbolically from an explicit objective, its sufficient supports, and session state. Work on query/result interestingness and exploratory querying asks complementary questions about novelty, surprise, exploration value, or exploratory interaction [11], [12]. MCAD does not treat interestingness as equivalent to strategic computability.

E. Intentional Analytics

Intentional Analytics raises the interaction level from individual operations to analytical intentions. The Intentional Analytics Model (IAM) models intentions and their operationalization over enriched multidimensional spaces [13]; more recent work explores intention prediction in this setting [14]. A particularly close neighbor is the *assess* operator of Francia *et al.*, which evaluates a target cube by comparing its observed performance with a reference cube or benchmark and then labeling the comparison outcome [27]. This is close to MCAD because a benchmark may encode expected KPI performance, but the evaluated property is different: *assess* judges an observed value against an expectation, whereas MCAD asks whether query-provided evidence completes sufficient supports and thereby changes which constraints of an explicit objective are computable relative to session state. MCAD is therefore compatible with this line but addresses a distinct runtime property once a query or plan exists.

F. Objectives, KPIs, and the Path to Multidimensional Models

Goal-oriented BI and decision-support approaches establish links among organizational objectives, KPIs, processes, and analytical requirements [15]. Semantic KPI models make indicator definitions, formulas, units, and dependencies explicit [16], [17], [18]. Requirements engineering for data warehouses also derives data needs and multidimensional models from strategic objectives [28]. MCAD therefore does not claim to invent the strategy–KPI–warehouse link. It extends this continuum to runtime, query by query: the active strategic objective is decomposed into constraints and sufficient supports, and the system evaluates which of those constraints each query actually makes computable in the current session state.

Multiple or conflicting objectives and state-dependent policies are natural extensions of the framework; contextual optimization is a relevant neighboring line for the latter [29]. The evaluated version of MCAD does not yet optimize conflicting objectives or learn an action policy.

G. Knowledge Graphs, Semantic OLAP, and Provenance

Knowledge graphs provide a general framework for representing entities, relations, schemas, and rules [19]; their automated construction and evolution remain active research topics [20]. In DW/BI, ontologies support semantic modeling, integration, and interoperability [30]. QB4OLAP and the RDF Data Cube Vocabulary connect semantic representation with multidimensional structures [31], [32]; KG-OLAP formalizes multidimensional operations over contextualized graphs [33], while Staudinger *et al.* use a graph to retain context and provenance during analysis [34].

Aggregate-result provenance and support justification:

Aggregate-result provenance provides a complementary neighborhood for explainability. AlOmeir *et al.* propose provenance summaries for aggregate queries to make the influence of source data more readable and to compare provenance across aggregate results [35]. This problem remains distinct from MCAD: provenance explains which data a result depends on; MCAD explains why supports associated with that result do, or do not, make an objective constraint computable. A provenance summary could therefore be attached to each realized support as additional justification without changing the definition of $\Delta\varphi$.

This integration is not evaluated as an independent component of the current prototype. Provenance justifies the dependence of a result on source data; by itself, it does not validate the business relevance of the constraint or the sufficiency of the support for the objective.

SHACL provides a standard graph-validation language [21], and PROV-O a standard provenance model [22]; recent why-provenance work further informs explanations for why an answer exists [36].

In MCAD, the CKG has a more specific responsibility: anchoring objective constraints and sufficient supports and maintaining the semantic memory required for session-relative contribution computation. Current inspectability is derived from symbolic objects—SAT clauses, realized supports, computable constraints, mask S^* , and marginal gain—rather than fine-grained source-tuple provenance. In summary, provenance explains the origin of evidence; symbolic traceability preserves the MCAD reasoning chain; MCAD determines the contribution of that evidence to objective computability.

H. Value Purpose and the Targeted Gap

The preceding families do not necessarily optimize the same object. Personalization and recommendation primarily target interaction adaptation and continuity; *interestingness* characterizes informational or cognitive interest; IAM formalizes analytical intention; benchmark-based assessment compares observed performance with a reference [27]; KPI models organize objective-indicator alignment; summarizability controls aggregation correctness; provenance explains the origin or reasons of a result. MCAD evaluates another property: verifiable progress in the computability of an explicit strategic objective, at each query and relative to session state. This distinction is a conceptual positioning axis, not a claim of psychological superiority or an additional experimental question.

The gap is therefore formulated cautiously. To the best of our knowledge within the reviewed corpus, these functions remain primarily distributed across specialized research lines. MCAD aims to articulate them around a common operational semantics: strategic objective $\rightarrow$ explicit constraints $\rightarrow$ sufficient supports $\rightarrow$ canonical plan $\rightarrow$ admissibility $\rightarrow$ realizable supports $\rightarrow$ computable constraints $\rightarrow$ marginal contribution, with session memory and a separate action policy. The distinctive contribution lies in this query/session-level unit of evaluation and in the chain that makes it computable and explainable.

I. Direct Comparison with the Closest Approaches

Table I provides a direct comparison with the neighbors that most clearly expose the novelty boundary: exploration contribution, session recommendation, context-aware OLAP recommendation, Intentional Analytics, and goal-oriented requirements engineering. The dimensions distinguish input and unit of analysis, state/context, evaluated purpose, computational mechanism, and output. Cells describe the center of gravity of the cited work rather than an exhaustive absence of capabilities.

The comparison is meant to locate MCAD, not to claim ownership of individual building blocks. Djedaini *et al.* learn

contribution within an exploration; Aligon *et al.* and Negre *et al.* help select or filter what comes next; IAM raises the level at which an analysis is expressed; RE4DW links strategic goals to multidimensional design; summarizability asks whether a computation is semantically sound; and provenance traces a result back to its source data. MCAD works at runtime between these levels: it turns an explicit objective into constraints and sufficient supports and asks how much the current query changes their computability given the session so far. In short: *summarizability asks whether the computation is correct; provenance asks where the result comes from; MCAD asks what the result makes newly computable for the objective.*

III. OVERVIEW OF MCAD

At the system level, MCAD sits between the analyst and the warehouse as a semantic decision layer. It asks two questions in sequence: is the query coherent with the decision-analysis context, and, if so, what does it add to the computability of the active objective?

A. Conceptual Chain from Query to Objective

Figure 2 summarizes the central conceptual chain: an OLAP query is compiled into a canonical query plan QP and then subjected to semantic mapping against the CKG, the current objective, and session state. This operation induces a contextual mask S_t^* whose evaluation characterizes marginal contribution before a configurable action policy is applied.

B. Main Components of MCAD

The contextual knowledge graph (CKG) is the operational representation of the **decision-analysis context**. It brings together the multidimensional catalog, KPI semantics, units and aggregators, business constraints, scenarios, temporal windows, the active objective, and the relations needed to interpret session state. In that sense, the CKG is more than a catalog: it is active semantic memory linking what the warehouse can provide with what the objective still needs in order to become computable.

The active strategic objective is modeled as a finite set of constraints attached to explicit KPI semantics. For each constraint, MCAD identifies the minimal analytical cells required for evaluation: combinations of fact, grain, measures, aggregators, units, filters, and period. These cells are represented as *virtual nodes*.

Each analytical query is reduced to a canonical query plan containing the semantics that matter to MCAD: target fact, grain, measures and aggregators, units, filters, and temporal window. This keeps the reasoning layer independent of surface syntax. The evaluated prototype includes extractors for MDX and analytical SQL; other languages would require an equally reliable mapping to QP .

Once a plan is contextually valid, MCAD determines which objective-related virtual nodes become realizable and then which constraints become computable. Semantic mapping among the plan, context, and supports induces an inspectable contextual mask. The mask itself chooses no action: its evaluation feeds the contribution measure, optionally weighted, and the action policy then selects the operational response.

Table I
DIFFERENTIATION MATRIX BETWEEN MCAD AND DIRECTLY RELEVANT RESEARCH LINES. CELLS DESCRIBE THE CENTER OF GRAVITY OF THE CITED WORK; THEY ARE NOT EXHAUSTIVE CLAIMS THAT OTHER CAPABILITIES ARE ABSENT.

Work / line	Input / unit	State / context	Purpose or criterion	Mechanism and output	Position relative to MCAD
Djedaini <i>et al.</i> [10]	query within an OLAP exploration	an exploration history	contribution to exploration quality/progression	query features, supervised learning, and competence tracking; automatic assessment	MCAD derives contribution from explicit constraints and realizable supports relative to session state rather than learning it from traces.
Aligon <i>et al.</i> [6]	current session + session log	session and historical neighborhood	relevance of a continuation	collaborative filtering, alignment, and adaptation; recommendation of a future session	MCAD evaluates the current query; its indicator can then feed a recommendation policy.
Negre, Ravat, Teste [7]	OLAP query or recommendation + context	recommendation context and obsolescence	avoid inappropriate or non-computable recommendations	obsolescence indicator + post-filtering; contextual recommendation	MCAD also uses context but first measures progress toward an explicit objective, then separates that measure from the action.
Intentional Analytics [13], [14]	analytical intention + enriched cube	session, previous results, and models	express a high-level analytical need	intentional operators and models; enriched answer / insight	IAM raises the level at which analysis is expressed; MCAD formalizes objective computability and its query-by-query variation.
Assessment / assessment operator [27]	target cube + reference benchmark	expected-vs-observed performance context	assess a measure relative to an expectation	benchmark comparison + labeling of the outcome	Close on KPI-oriented assessment, but MCAD evaluates support completion and session-relative objective computability rather than deviation from an expected value.
RE4DW [28]	strategic objectives and KPIs	DW design context	derive data requirements and a multidimensional model	requirements engineering; requirements and multidimensional model	RE4DW links strategy and design; MCAD extends that continuum to runtime through sufficient supports and marginal gain.
Summarizability / semantic validation [25]	measure, dimension, grain, aggregation	schema, hierarchies, summarizability properties	semantic correctness of the computation	summarizability conditions and propagation; valid/invalid aggregation	A useful condition for MCAD admissibility, but not equivalent to support realization or contribution.
Aggregate provenance [35]	aggregate result + source data	computation dependencies and data origin	explain data origin and influence	provenance summaries; result dependencies	Provenance explains the evidence; MCAD determines whether the associated support makes a constraint computable.
MCAD	QP_t , objective CKG	O , decision-analysis context + H_t	complete/partial computability and marginal gain	$SAT / Real / C_{eval} / \varphi, \Delta\varphi$; indicator then configurable policy	unit of evaluation: what the query makes newly computable for the objective given the session state.

Note— Neighboring methods were not reimplemented as competing baselines in the reported campaigns; this matrix establishes conceptual and mechanistic differentiation, not direct experimental superiority.

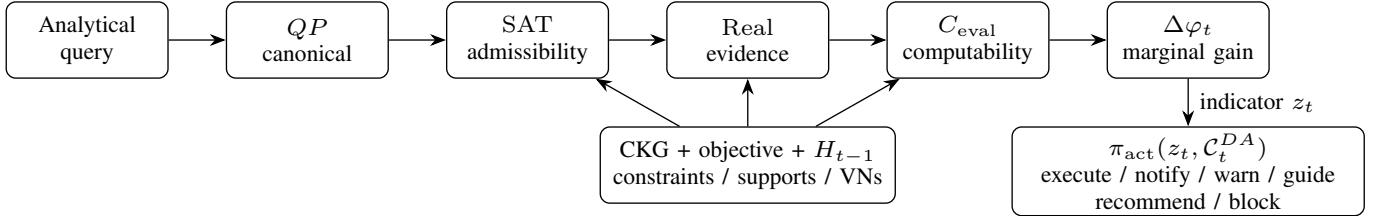


Figure 2. Central conceptual chain of MCAD: from the query to the strategic-computability indicator and then to a configurable action policy.

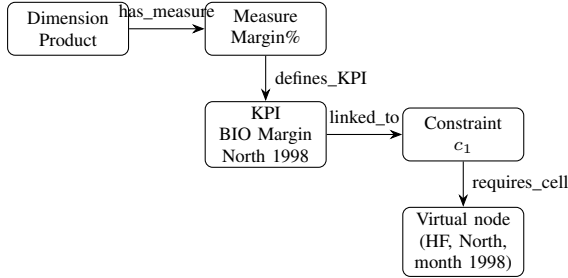


Figure 3. Simplified CKG fragment linking a dimension, a measure, a KPI, a constraint, and a virtual node.

C. Illustrative CKG Fragment and Practical Construction

Figure 3 shows a simplified CKG fragment centered on a dimension, a measure, a KPI, a constraint, and a virtual node.

In practice, building the CKG draws on several sources: the multidimensional schema, technical metadata such as data dictionaries and ETL rules, business documentation such as KPI definitions and reporting procedures, and expert workshops

used to surface rules that are otherwise implicit—for example incompatible filters, scenario assumptions, admissible units, or specific temporal windows.

A deployment could bootstrap the CKG from schemas and metadata and enrich it with business knowledge, but the present experiments use CKGs prepared from controlled schemas, rules, and objectives. Automatic construction from documents, logs, or implicit rules therefore remains future work. The evidence concerns reasoning over a specified CKG, not automatic knowledge acquisition; the graph could be stored in a property-graph database or RDF store.

IV. FORMALIZATION OF MCAD

This section introduces the main formal objects of MCAD in the order in which they are needed by the reasoning chain. We first distinguish the multidimensional schema from the *decision-analysis context* and then show how that context is represented and queried in the CKG. We next define strategic objectives, virtual nodes, canonical query plans, contextual validity, contribution measures, and the action policy. Detailed proofs are provided in Appendix A.

Table II
MAIN NOTATION OF THE MCAD MODEL.

Symbol	Description
F	Set of fact tables
D	Set of dimensions
$L(D)$	Levels of dimension D
g	Analysis grain (product of levels)
M	Set of measures and KPIs
CKG	Contextual knowledge graph
O	Active strategic objective
$C^*(O)$	Set of constraints of O
$N(c)$	Virtual nodes associated with constraint c
QP	Canonical query plan
$\text{Real}(QP)$	Virtual nodes realizable by QP
$C_{\text{eval}}(QP, O)$	Constraints fully computable via QP
E_t	Virtual-node evidence acquired and retained by session step t
$C_Q^{\leq t}(O)$	Query-wise accumulation of individually completed constraints
$C_E^{\leq t}(O)$	Evidence-wise constraints computable from cumulative evidence E_t
$\kappa_c(X)$	Intra-constraint computability degree of c
$\psi(QP, O)$	Graded computability induced by QP
$\varphi(QP, O)$	Coverage of fully computable constraints

Importantly, the contextual mask should not be conflated with the CKG, with C_{eval} , or with $\Delta\varphi_t$. The CKG provides the semantic reference frame, C_{eval} denotes computable constraints, and $\Delta\varphi_t$ measures marginal computability gain. It is the semantic representation of the mapping among the query plan, the context, the supports, and the objective constraints.

A. Decision-Analysis Context and Contextual Knowledge Graph

We consider a warehouse organized according to a classical multidimensional schema [1], [2], with a set of fact tables F , a set of dimensions D organized into level hierarchies, and a set of measures M . For a dimension $D \in D$, let $L(D)$ denote its hierarchical levels. An *analysis grain* is a Cartesian product of levels:

$$g = \prod_{D \in D_g} \ell_D, \quad \ell_D \in L(D),$$

where $D_g \subseteq D$ is the set of dimensions represented in grain g .

An *elementary analytical entity* is a triplet (f, g, m) , where $f \in F$ is a fact table, g is a grain, and m is a measure or derived KPI computed at that grain.

Table II summarizes the main notation used throughout the paper.

The decision-analysis context at step t is represented conceptually as

$$C_t^{DA} = (W, O, C^*(O), K, \Sigma, U, \Omega, H_t),$$

where W denotes the multidimensional warehouse/catalog context, O the active strategic objective, $C^*(O)$ its constraints, K KPI and business semantics, Σ contextual and summarizability rules, U unit semantics, Ω temporal/scenario information, and H_t the cumulative session state. This complete context is operationalized in a contextual knowledge graph

$$\text{CKG} = (V, E, L),$$

where V is a set of nodes, $E \subseteq V \times L \times V$ a set of labeled edges, and L the relation types. Nodes include schema elements,

computational objects, business constraints, scenarios, temporal windows, objective elements, and session-state objects.

Knowledge-graph, semantic-indicator, and multidimensional-design research motivates graph representations and formal analytical objects [19], [18], [37]. In MCAD, relation names such as *rolls_up_to*, *has_measure*, *defines_KPI*, *uses_unit*, *allowed_agg*, *requires_grain*, *conflicts_with*, and *applicable_in_scenario* are prototype design choices and are not attributed to the cited literature. KPI computation rules and business constraints are encoded as additional nodes and edges.

B. Strategic Objectives, Constraints, and Virtual Nodes

A strategic objective is modeled as

$$O = (K^*(O), C^*(O)),$$

where $K^*(O)$ is the set of KPIs relevant to the objective and $C^*(O)$ is a finite set of analytical constraints. Each constraint $c \in C^*(O)$ is associated with a KPI $\text{KPI}(c) \in K^*(O)$ and a predicate r_c that may return either a Boolean value or a degree of satisfaction in $[0, 1]$ once the required data are available. MCAD explicitly distinguishes *computability* of a constraint—whether the quantity required for evaluating it can be computed—from its *actual satisfaction* on data.

To connect constraints with warehouse structure, MCAD introduces *virtual nodes*

$$nv = (f, g, m, \alpha, u, S, w),$$

where $f \in F$ is a fact table, g a grain, $m \in M$ a measure or KPI, α an aggregation operator, u a unit, S a set of filters, and w a temporal window. A virtual node therefore describes a minimal analytical cell characterized by a cube position and an observed or derived quantity.

For each constraint $c \in C^*(O)$, let $\mathcal{R}(c)$ denote a finite nonempty family of sufficient supports, each support being a finite nonempty set of virtual nodes. Throughout this formal section, $C^*(O)$ is finite and nonempty; for every $c \in C^*(O)$, $1 \leq |\mathcal{R}(c)| < \infty$; and every $R \in \mathcal{R}(c)$ is finite and nonempty. The relation “sufficient support” is an input of the objective model and the CKG; it is not automatically inferred by the evaluated MCAD reasoner. MCAD checks realization of the specified supports and derives the corresponding computability. The business validity of that specification is therefore an assumption of the mechanism.

Illustrative virtual node: Consider a virtual node associated with a margin constraint for a BIO product family in the North region:

$$nv_1 = (\text{SALES}, g, \text{Margin}\%, \text{avg}, \%, S, w),$$

where g fixes the grain $\{\text{Month}=2024-03, \text{ProductFamily}=BIO, \text{Region}=North\}$, $S = \{\text{Channel} = \text{Physical Store}\}$, and $w = [2024-03-01, 2024-03-31]$. A measured average margin of 18.7% for that cell can therefore be represented independently of the query language used to obtain it.

Table III
EXAMPLES OF SLICER RELATIONSHIPS IN THE CKG.

Relation	Interpretation	Example
mutex	Forbidden combination	Channel=Online & Store=Physical
includes	B includes A	Region=North includes City=Lille
coherent	Allowed combination	Family=BIO & Promo=Yes

C. Canonical Query Plans and Contextual Validity

Each analytical query is abstracted as a *canonical query plan*

$$QP = (f_Q, g_Q, M_Q, \alpha_Q, u_Q, S_Q, w_Q),$$

where $f_Q \in F$ is the target fact table, g_Q is the canonical query grain, $M_Q \subseteq M$ the requested measures or KPIs, α_Q maps measures to aggregation/computation operators, u_Q maps measures to semantic units, S_Q is a finite set of normalized atomic slicers, and w_Q is a canonical description of the temporal component. Grain g_Q is defined as a partial mapping from active dimensions to hierarchical levels after normalization through the CKG.

The plan is not raw query syntax but a semantic canonical representation produced by an extraction operator

$$\mathcal{E}_L(q, CKG) = QP,$$

where q is written in an analytical language L such as MDX or analytical SQL. Presentation-only differences that induce the same analytical selection therefore lead to the same canonical plan.

Contextual validity is captured by

$$\begin{aligned} \text{SAT}(QP) \Leftrightarrow & \text{grain_ok}(QP) \wedge \text{agg_ok}(QP) \\ & \wedge \text{unit_ok}(QP) \wedge \text{slc_ok}(QP) \\ & \wedge \text{time_ok}(QP) \wedge \text{nvac_ok}(QP). \end{aligned} \quad (1)$$

Each clause represents a family of constraints derived from the CKG:

- grain_ok checks grain compatibility with hierarchies and summarizability rules;
- agg_ok checks whether requested aggregation operators are semantically authorized for the measures and grain; this check concerns computation correctness, not contribution;
- unit_ok checks unit consistency;
- slc_ok checks non-contradiction and filter compatibility;
- time_ok checks temporal consistency; and
- nvac_ok checks that the queried subspace is not *a priori* empty according to the CKG.

Relationships among slicers such as mutex, includes, and coherent are especially important for slc_ok. Table III illustrates representative cases.

Algorithm 1 gives a simplified computation of SAT(QP).

D. Realizable Evidence, Computable Constraints, and the Contextual Mask

In the operational pipeline, SAT(QP) is the first safeguard. If SAT(QP) = false, the strict policy does not authorize the

Algorithm 1 Contextual-validity check SAT(QP)

Require: Query plan QP , graph CKG

Ensure: $\text{SAT}(QP) \in \{\text{true}, \text{false}\}$

```

1: Extract  $(f_Q, g_Q, M_Q, \alpha_Q, u_Q, S_Q, w_Q)$  from  $QP$ 
2: if grain_ok( $QP, CKG$ ) is false then
3:   return false
4: end if
5: if agg_ok( $QP, CKG$ ) is false then
6:   return false
7: end if
8: if unit_ok( $QP, CKG$ ) is false then
9:   return false
10: end if
11: if slc_ok( $QP, CKG$ ) is false then
12:   return false
13: end if
14: if time_ok( $QP, CKG$ ) is false then
15:   return false
16: end if
17: if nvac_ok( $QP, CKG$ ) is false then
18:   return false
19: end if
20: return true

```

query and Real and C_{eval} are not used to justify authorization. Conversely, $\text{SAT}(QP) = \text{true}$ does not imply positive contribution: an admissible query may still have $\Delta\varphi_t = 0$.

Given objective O and plan QP , let

$$\text{Real}(QP) \subseteq N(O)$$

denote the virtual nodes realizable by QP . A virtual node $nv = (f, g, m, \alpha, u, S, w)$ is realizable if the fact, grain, measure, aggregator, unit, slicers, and temporal window are canonically compatible with the query plan. Formally,

$$\text{Real}(QP) = \{nv \in N(O) \mid \text{Realizable}(nv, QP)\}.$$

Thus, Real(QP) denotes fully compatible virtual-node evidence rather than the mere presence of a measure in the query.

For each $c \in C^*(O)$, let

$$\mathcal{R}(c) = \{R_{c,1}, \dots, R_{c,p_c}\}, \quad R_{c,j} \subseteq N(O),$$

where each $R_{c,j}$ is sufficient to make c computable. We define

$$C_{\text{eval}}(QP, O) = \{c \in C^*(O) \mid \exists R \in \mathcal{R}(c), R \subseteq \text{Real}(QP)\}.$$

The existential quantifier is essential: complete computability of a constraint is obtained as soon as at least one sufficient support is fully realized; alternative supports need not be realized simultaneously. If each constraint has a unique support $N(c)$, then

$$C_{\text{eval}}(QP, O) = \{c \in C^*(O) \mid N(c) \subseteq \text{Real}(QP)\}.$$

Graded intra-constraint computability: The preceding definition characterizes complete constraint computability. To represent internal progress before a sufficient support is fully realized, define support coverage for $R \in \mathcal{R}(c)$ and $X \subseteq N(O)$ as

$$\rho(R, X) = \frac{|R \cap X|}{|R|} \in [0, 1], \quad R \neq \emptyset.$$

The intra-constraint computability degree is

$$\kappa_c(X) = \max_{R \in \mathcal{R}(c)} \rho(R, X).$$

Hence $\kappa_c(X) = 0$ means that no element of a sufficient support is yet realized, $0 < \kappa_c(X) < 1$ represents partial intra-constraint computability, and $\kappa_c(X) = 1$ means that at least one sufficient support is complete. For a query,

$$\psi(QP, O) = \frac{1}{|C^*(O)|} \sum_{c \in C^*(O)} \kappa_c(\text{Real}(QP)).$$

A constraint belongs to $C_{\text{eval}}(QP, O)$ if and only if $\kappa_c(\text{Real}(QP)) = 1$; φ therefore counts only fully computable constraints, whereas ψ preserves intra-constraint progress.

At session level, let $E_0 \subseteq N(O)$ denote evidence already acquired and retained at the beginning of a semantic phase. For $t \geq 1$, let A_t contain only evidence actually acquired at step t , with $A_t \subseteq \text{Real}(QP_t)$ and monotone update $E_t = E_{t-1} \cup A_t$. A blocked, non-executed, or otherwise non-acquired candidate has $A_t = \emptyset$. Define

$$\begin{aligned} \psi^{\leq t}(O) &= \frac{1}{|C^*(O)|} \sum_{c \in C^*(O)} \kappa_c(E_t), \\ \Delta\psi_t &= \psi^{\leq t}(O) - \psi^{\leq t-1}(O). \end{aligned}$$

The reported campaigns, however, use the binary $\varphi/\Delta\varphi$ measure based on fully computable constraints; $\psi/\Delta\psi$ is a formalized extension that is not experimentally evaluated in the present study and is not used to reinterpret the existing ALLOW/BLOCK decisions.

Phase assumptions and interpretation conventions: The formal properties are conditional on a fixed *semantic phase*. (H1) Objective O , constraints $C^*(O)$, sufficient supports $\mathcal{R}(c)$, and contextual rules remain fixed within a phase; an explicit change in KPI, scope, business rule, or support defines a new comparison baseline. (H2) Support elements are canonical, semantically atomic analytical units at the chosen representation level and are not arbitrarily refined within the phase. (H3) With alternative supports, the maximum in κ_c measures progress toward the currently most advanced sufficient alternative; it is neither a probability of success nor a business-satisfaction degree. (H4) Only evidence actually acquired in E_t is counted; unknown or unobserved components are not treated as realized. (H5) In the weighted variant, weights are positive, exogenous priorities fixed with the objective and are not learned from MCAD decisions. The experiments in this paper evaluate the unweighted $\varphi/\Delta\varphi$ version.

Semantic mapping is denoted conceptually by $\mathcal{M}(QP, C_t^{DA}, O)$. The induced contextual mask S_t^* represents canonical-plan elements retained by this mapping with the context and objective supports. In the evaluated formal instantiation,

$$S^*(QP, O) = \bigcup_{c \in C_{\text{eval}}(QP, O)} \bigcup_{\substack{R \in \mathcal{R}(c) \\ R \subseteq \text{Real}(QP)}} R,$$

and, in the unique-support case,

$$S^*(QP, O) = \bigcup_{c \in C_{\text{eval}}(QP, O)} N(c).$$

The mask S_t^* does not necessarily represent all query results and replaces neither computable constraints nor the contribution measure. In the prototype, mask materialization is an inspectable object linked to the CKG and session state; it is neither a rewrite of the multidimensional catalog nor fine-grained source-tuple provenance.

The following chain is a complementary conceptual view and does not modify the preceding equations:

$$\begin{aligned} q_t &\rightarrow QP_t \rightarrow \mathcal{M}(QP_t, C_t^{DA}, O) \rightarrow S_t^* \\ &\rightarrow \text{Eval}(S_t^*) \rightarrow \Delta\varphi_t \rightarrow \pi_{\text{act}}. \end{aligned}$$

Here $\mathcal{M}$ is semantic mapping, S_t^* the contextual mask, $\text{Eval}(S_t^*)$ contribution evaluation, $\Delta\varphi_t$ marginal contribution, and π_{act} the action policy.

E. Contribution Measures, Session Progress, and Complexity

For $|C^*(O)| \geq 1$, the unweighted contribution of QP is

$$\varphi(QP, O) = \frac{|C_{\text{eval}}(QP, O)|}{|C^*(O)|} \in [0, 1].$$

A weighted variant uses positive weights $w_c > 0$:

$$\varphi_w(QP, O) = \frac{\sum_{c \in C_{\text{eval}}(QP, O)} w_c}{\sum_{c \in C^*(O)} w_c}.$$

Weights express exogenous business priorities and remain fixed within a phase; modifying them is an explicit change in priorities. The experimental results below use the unweighted measure.

For session semantics, two cumulative sets must be distinguished. Evidence-wise accumulation is canonical:

$$C_E^{\leq t}(O) = \{c \in C^*(O) \mid \exists R \in \mathcal{R}(c), R \subseteq E_t\}.$$

For comparison with query-wise completion while allowing a nonempty initial state, define

$$C_Q^{\leq t}(O; E_0) = C_E^{\leq 0}(O) \cup \bigcup_{i=1}^t C_{\text{eval}}(QP_i, O).$$

When $E_0 = \emptyset$, this reduces to the query-wise union exercised by the audited historical sessions. We define

$$C_{\text{eval}}^{\leq t}(O) := C_E^{\leq t}(O), \quad \varphi^{\leq t}(O) = \frac{|C_{\text{eval}}^{\leq t}(O)|}{|C^*(O)|},$$

and $\Delta\varphi_t = \varphi^{\leq t}(O) - \varphi^{\leq t-1}(O)$.

Two per-step conditions make the query-wise shortcut testable. (A1) *Query-to-acquisition consistency*: $C_{\text{eval}}(QP_i, O) \subseteq C_E^{\leq i}(O)$. (A2) *Single-query completion of newly computable constraints*: $C_E^{\leq i}(O) \setminus C_E^{\leq i-1}(O) \subseteq C_{\text{eval}}(QP_i, O)$. A2 means that whenever a constraint first becomes evidence-wise computable, the current query alone realizes a complete sufficient support; it does not require singleton supports. The audited Q1–Q6, B/C, and frozen RQ4 paths satisfy these conditions transition by transition. For FoodMart-1000, the 2266 acquired ALLOW events and their cumulative progression are auditable, but raw C_{eval} outputs for the 5694 blocked candidates were not retained; therefore exhaustive per-candidate equality over all

7960 decisions is not claimed. No published numerical result is changed by this boundary clarification.

Two kinds of partiality must be distinguished. At objective level, coverage is null when $C_{\text{eval}} = \emptyset$, partial when only a proper subset of $C^*(O)$ is computable, and complete when $C_{\text{eval}} = C^*(O)$. At individual-constraint level, $0 < \kappa_c(X) < 1$ instead represents partial computability within one constraint. Contribution is query- and session-relative. Reproducing already acquired evidence can give $\Delta\varphi_t = 0$ and $\Delta\psi_t = 0$, but $\Delta\varphi_t = 0$ does not generally imply $A_t = \emptyset$ or $\Delta\psi_t = 0$: under a non-strict policy, an executed query may acquire authorized partial evidence without yet completing a sufficient support. The reported strict campaigns use the binary complete-support measure and are not reinterpreted through this graded case.

Proposition 1. *For every objective O and query plan QP with $|C^*(O)| \geq 1$, $\varphi(QP, O) \in [0, 1]$. Moreover, $\varphi(QP, O) = 0$ iff $C_{\text{eval}}(QP, O) = \emptyset$, and $\varphi(QP, O) = 1$ iff $C_{\text{eval}}(QP, O) = C^*(O)$.*

Proposition 2. *If $\text{Real}(QP_a) \subseteq \text{Real}(QP_b)$, then $\varphi(QP_a, O) \leq \varphi(QP_b, O)$.*

Proposition 3. *For session $(QP_1, \dots, QP_T)$, $(\varphi^{\leq t}(O))$ is monotone non-decreasing and bounded in $[0, 1]$; in particular $\Delta\varphi_t \geq 0$.*

Proposition 4. *For every query and session state, $\kappa_c \in [0, 1]$ and $\psi \in [0, 1]$. Moreover $\varphi(QP, O) \leq \psi(QP, O)$; under monotone evidence accumulation $E_{t-1} \subseteq E_t$, $\psi^{\leq t-1}(O) \leq \psi^{\leq t}(O)$ and $\Delta\psi_t \geq 0$.*

Proposition 5. *Assume, within one fixed semantic phase, that for every step i : (A1) $C_{\text{eval}}(QP_i, O) \subseteq C_E^{\leq i}(O)$ and (A2) $C_E^{\leq i}(O) \setminus C_E^{\leq i-1}(O) \subseteq C_{\text{eval}}(QP_i, O)$. Then, for every t , $C_Q^{\leq t}(O; E_0) = C_E^{\leq t}(O)$. If (A1) is retained but (A2) is dropped, then $C_Q^{\leq t}(O; E_0) \subseteq C_E^{\leq t}(O)$, and the inclusion can be strict.*

With indexed virtual-node/support representations and constant-time indexed membership/compatibility lookups, a direct scan of the relevant reasoning structures has core cost

$$O\left(|N_{\text{cand}}(O)| + \sum_{c \in C^*(O)} \sum_{R \in \mathcal{R}(c)} |R|\right),$$

that is, linear in the candidate virtual nodes and support elements actually traversed under these assumptions. This bound concerns the reasoning core after indexed structures have been built. It excludes canonical-plan extraction, CKG construction or maintenance, backend data access, and resolution of general non-indexed business rules; it is therefore not an end-to-end complexity bound for the analytical system.

F. Contribution Indicator and Action Policy

The scientific output of MCAD is a contribution evaluation, not an intrinsic ALLOW/BLOCK decision. Let

$$z_t = (\text{SAT}(QP_t), C_{\text{eval}}(QP_t, O), S^*(QP_t, O), \varphi^{\leq t}(O), \Delta\varphi_t)$$

collect the relevant formal objects. A configurable policy

$$\pi_{\text{act}}(z_t, C_t^{DA}) \in \{\text{EXECUTE}, \text{NOTIFY}, \text{WARN}, \text{GUIDE}, \text{RECOMMEND}, \text{BLOCK}\}.$$

selects the operational action. A *non-contributive query* denotes an admissible query whose marginal computability gain is zero relative to the current objective and session state; it does not mean that the query is syntactically invalid or useless to a user or another analysis. The strict experimental policy maps inadmissible or non-contributive cases to BLOCK; this mapping belongs to the protocol and is not part of the scientific definition of contribution.

G. Canonical Abstraction and Query-Language Independence

MCAD depends on canonical representation $QP = (f_Q, g_Q, M_Q, \alpha_Q, u_Q, S_Q, w_Q)$ rather than on a particular query syntax. For any analytical language L with a reliable extractor $\mathcal{E}_L(q, CKG) = QP$, the same SAT/Real/ C_{eval} / φ reasoning core can in principle be reused. This interface property does not imply equally mature extractors for all languages. The prototype paths and the experimentally exercised backends are bounded explicitly below.

V. PROTOTYPE ARCHITECTURE AND WORKFLOW

The prototype separates the CKG-based decision-analysis context, MCAD contribution reasoning, action policy, and analytical execution so that the computability model is not conflated with the strict ALLOW/BLOCK policy used in some experiments. Figure 4 shows these layers.

A. Session Layer and Query Acquisition

The analyst works within a session associated with an active strategic objective. The interface maintains the objective identity O , the current query q_t , and session state H_{t-1} . Input adapters extract a canonical query plan QP_t from MDX or analytical SQL—or directly receive one. The canonical representation lets the reasoner work over homogeneous semantics independently of the implemented front-end syntax.

B. Decision-Analysis Context / CKG Layer

The CKG is organized around four families of objects: (i) the multidimensional catalog (facts, dimensions, hierarchies, measures); (ii) business semantics (KPIs, units, aggregators, scenarios, and temporal rules); (iii) the objective model (constraints $C^*(O)$, virtual nodes, and sufficient supports); and (iv) session memory (already computable constraints and observed useful supports). This layer provides SAT, Real, and C_{eval} with the relations required to reason over the decision-analysis context rather than over query syntax alone.

C. MCAD Reasoning Core

The semantic pipeline preserves the formal chain $QP \rightarrow \text{SAT} \rightarrow \text{Real} \rightarrow C_{\text{eval}} \rightarrow S^* \rightarrow \varphi, \Delta\varphi$. Conceptually, the CKG is the reference frame in which the plan is semantically mapped; the contextual mask is the induced representation of that mapping for the query, objective, and current session state. Mask evaluation feeds the contribution measure, after which policy π_{act} converts that measure into an operational action. Outputs are retained as inspectable objects: admissibility

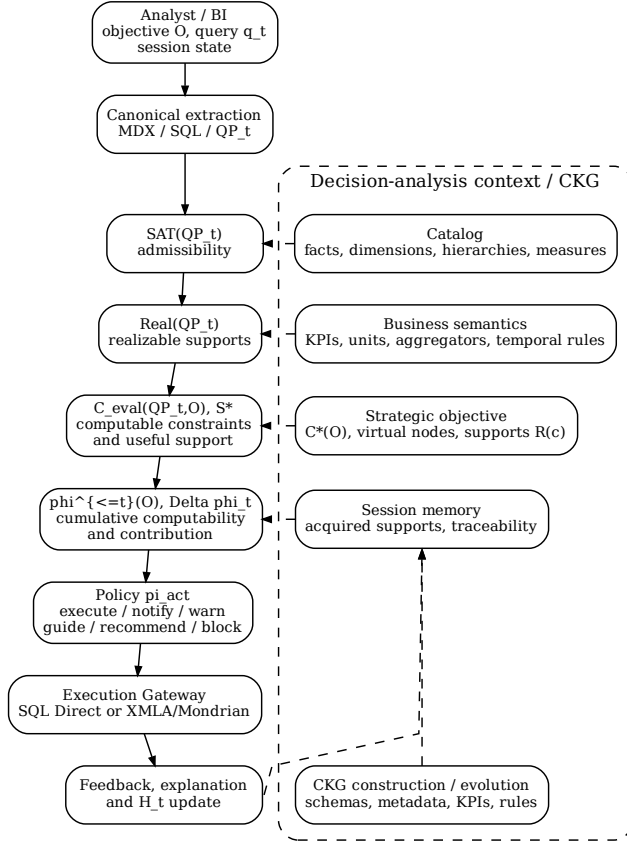


Figure 4. MCAD single-column architecture. The CKG materializes the decision-analysis context; the MCAD core computes admissibility, realizable supports, computable constraints, and marginal contribution; a configurable action policy then turns that evaluation into interaction or control before any physical execution.

clauses, realized/missing supports, computable constraints, contextual mask, and cumulative progress. The evaluated prototype implements and exposes the query-level reasoning chain and the session behavior exercised by the frozen historical campaigns. The generalized evidence-wise C_E semantics—including multi-query support assembly and arbitrary alternative-support accumulation—is formally specified and regression-tested by the reference model, but this paper does not claim that every such generalized case is fully implemented by the separately inspected candidate session accumulator.

D. Action Policy and Physical Execution

The reasoning output is sent to π_{act} , which selects an action according to deployment context. An assistance policy can show a warning or notification for a redundant query; a guidance policy can suggest a query with expected $\Delta\varphi > 0$; a recommendation policy can rank alternatives; a permissive policy can execute despite $\Delta\varphi = 0$ while recording the contribution indicator; and a strict policy can use BLOCK to prevent physical effect. The demonstration prototype uses the latter configuration to verify that identified non-contribution can in fact be enforced before SQL Direct or XMLA/Mondrian execution.

E. User Feedback, Session Memory, and Traceability

The interface can return the result, newly computable constraints, $\varphi^{\leq t}(O)$, $\Delta\varphi_t$, and a formal explanation. On the evaluated paths, session state is updated with evidence credited by the protocol; the generalized E_t/C_E semantics remains the normative reference for future multi-query accumulation, while the prototype claim is limited to exercised paths. Traceability retains the canonical plan, relevant clauses, supports, mask, and action. Fine-grained tuple provenance is not implemented, although support-level summaries could be added without changing C_{eval} or $\Delta\varphi$.

F. End-to-End Workflow

The workflow is summarized as follows:

- 1) select or define objective O and initialize its context in the CKG;
- 2) submit query q_t and extract canonical plan QP_t ;
- 3) perform semantic mapping, construct mask S_t^* , and evaluate SAT, Real, C_{eval} , $\varphi^{\leq t}$, and $\Delta\varphi_t$;
- 4) apply the deployment-specific policy π_{act} (execution, notification, warning, guidance, recommendation, or blocking);
- 5) if the action requires it, forward the query to the backend and retrieve the result; and
- 6) update H_t and return explanation and session progress to the analyst.

VI. EXPERIMENTAL EVALUATION

A. Evaluation Questions and Protocol

The workloads come with an explicit reference specification: objectives, constraints, and sufficient supports are fixed before the tested policy runs. This lets us evaluate whether the reasoner behaves consistently with a given CKG and objective, but it does not independently establish that the decomposition is the best representation of business practice. We do not measure inter-expert agreement or agreement between formal computability and utility judged by practicing analysts. In the robustness benchmark, `oracle_allow` and `oracle_ceval` are part of the scenario specification and are read before the policy under test is invoked; they are not inferred from the policy’s own output.

The evaluation is organized around six research questions separating scientific contribution evaluation, operational policy instantiation, and prototype properties:

- **RQ1 – Contribution discrimination:** Does MCAD distinguish contributive, inadmissible, and redundant queries and correctly update cumulative session computability?
- **RQ2 – Pre-backend application:** When a strict ALLOW/BLOCK policy uses the MCAD indicator, is the action consistently applied before physical execution over a large number of sessions?
- **RQ3 – Semantic preservation:** Are decisions and their reasons preserved across several warehouses and between SQL Direct and XMLA paths in paired scenarios?
- **RQ4 – Controlled robustness and component sensitivity:** How does MCAD compare with internal reference

Table IV
COVERAGE OF THE MATHEMATICAL CORE BY THE EVALUATED PROTOTYPE AND THE V8.7.6 FORMAL REFERENCE MODEL.

Formal construction	Role in the model	Verified/evaluated realization	Observable evidence
Canonical plan QP	Language-independent analytical abstraction	MDX parser, analytical-SQL parser, and direct QP endpoint with canonical normalization on the evaluated paths.	<code>/parse_mdx</code> , <code>/parse_sql</code> , <code>/eval_qp</code>
$SAT(QP)$	Contextual admissibility	Named objective-sensitive clauses (grain, aggregation, unit, slicer, time, nonempty subspace) exercised by the prototype.	Clause-level explanations
$Real(QP)$	Realizable virtual-node evidence	Canonical compatibility on fact, grain, measure, aggregator, unit, slicers, and temporal window on the evaluated paths.	Realized-node identifiers and traces
$C_{eval}(QP, O)$	Fully computable constraints	Complete sufficient supports represented by requirement sets; query-level complete-support logic exercised by the frozen campaigns.	Computable constraints and selected complete support
$S^*(QP, O)$	Contextual mask	Inspectable mask induced from realized complete supports on the evaluated paths.	<code>induced_mask_node_ids</code> , <code>induced_mask_constraints</code> , <code>missing requirements</code>
$\varphi(QP, O)$ and historical session variants	Binary contribution and cumulative progress	Query-level φ and the session behavior exercised by Q1–Q6, B/C, FoodMart retained ALLOW evidence, and frozen RQ4 are supported by historical artifacts. The generalized evidence-wise E_t/C_E semantics, including multi-query support assembly and arbitrary alternative-support accumulation, is specified and regression-tested by the independent formal reference model; full implementation by the separately inspected candidate accumulator is not claimed.	<code>phi</code> , <code>phi_weighted</code> , <code>historical phi_leq_t/delta_phi_t</code> ; A2 <code>property/model-check</code> outputs

policies and targeted ablations under adversarial, noisy, and longer scenarios?

- **RQ5 – Structural growth:** How do CKG nodes, edges, and virtual nodes evolve as problem size increases?
- **RQ6 – Semantic-decision cost:** How do p50, p95, and p99 evolve as the number of constraints, number of virtual nodes, and membership density vary?

Table V shows how the six questions are covered. We deliberately keep three kinds of evidence separate: end-to-end physical evidence, where an authorized query actually reaches SQL Direct or XMLA; a controlled semantic-core benchmark using internal reference policies and targeted ablations; and structural/timing characterization of the MCAD layer itself. Using all three gives a broader empirical picture, but they are not interchangeable replications of the same experiment.

Metrics follow that separation: computability coverage and gain, consistency between selected action and physical effect, *false ALLOW/false BLOCK* rates, non-contributive executions, CKG size, and p50/p95/p99 semantic-decision quantiles. These quantiles do not measure SQL/XMLA execution time. The Q1–Q6 trace is one instrumented end-to-end sequence, not six independent replications; the FoodMart study contains 1000 sessions and 7960 decisions.

B. End-to-End Functioning Evidence

A six-query AdventureWorks trace illustrates the strict demonstration policy. Q1–Q3 successively provide supports for *SalesAmount*, *TotalProductCost*, and *GrossMargin*, moving cumulative computability from 1/3 to 2/3 to 1; each is authorized and physically executed. Q4–Q6 test out-of-objective scope, grain mismatch, and post-coverage redundancy and are stopped before the backend. The same contribution assessments could instead drive warning, guidance, or recommendation.

This scenario illustrates two core properties: contextual admissibility is necessary but not sufficient, and contribution is relative to session state. Q6 satisfies the validity clauses but adds no newly computable constraint; its marginal gain is zero. The strict demonstration policy therefore chooses BLOCK; that operational choice does not follow from $\Delta\varphi = 0$ alone.

C. Large-Scale Execution and Multi-Configuration Preservation

The FoodMart workload contains 1000 analytical sessions and 7960 decisions. Under the strict policy evaluated here, 2266 queries are authorized and 5694 are blocked. Every authorized query reaches physical execution, and every blocked query is stopped before the backend. Across the full workload, all 7960 decisions satisfy the canonical decision–physical-effect contract; no violation is observed. The A1 historical audit independently reconstructs the 2266 acquired ALLOW events and their cumulative progression. However, the campaign did not retain raw C_{eval} outputs for the 5694 blocked candidates; consequently, this paper does not claim an exhaustive per-candidate reconstruction of $C_Q^{\leq t} = C_E^{\leq t}$ over all 7960 decisions. This is a trace-retention boundary, not a change to the reported action or physical-execution counts.

A secondary reaggregation of recorded reasons, without new experimental runs, shows `BLOCK_REDUNDANT_DPHE_ZERO` for 3372 decisions: 42.4% of all decisions and 59.2% of BLOCK decisions. The seven mutually assigned primary-reason categories sum to all 7960 decisions, so this count denotes the explicit session-redundancy reason rather than every numerical occurrence of $\Delta\varphi = 0$. It shows that, under this strict policy, a large share of stops reflects no newly computable constraint; it is not a measure of human blocking utility.

The multi-warehouse study combines FoodMart/XMLA with AdventureWorks and SteelWheels/SQL Direct: 7 of 18 queries are executed and 11 stopped before the backend. A paired AdventureWorks study runs six logically equivalent queries through SQL Direct and XMLA with unchanged order, decisions, reasons, and execution policy. This demonstrates preservation only across the warehouses, adapters, and paths exercised, not universal portability or backend-performance equivalence.

D. Controlled Robustness and Targeted Ablations

Robustness is tested across four scenario families: adversarial admissibility cases, adversarial semantic cases, noisy borderline cases, and longer sequences. We compare MCAD

Table V
CLAIM–EVIDENCE–BOUNDARY MATRIX FOR THE EVALUATION: SUPPORTED SCOPE AND INFERENCE LIMIT FOR EACH EXISTING PROTOCOL.

RQ	Evidence / volume	What the evidence establishes	Interpretation boundary
RQ1	Physical Q1–Q6 trace; AdventureWorks SQL Direct; 1 session / 6 queries	Separation of contribution / inadmissibility / redundancy and cumulative updating on the instrumented trace.	Statistical replication, independent business validity of the objective, or user benefit.
RQ2	FoodMart XMLA/Mondrian; 1000 sessions / 7960 decisions	Repeated integrity of the decision → physical-effect contract under the strict policy.	Superiority of BLOCK as an interaction or independent validation of the business oracle.
RQ3	3 warehouses; 18 multi-DW queries + 12 paired SQL/XMLA evaluations	Preservation of decision semantics across the tested warehouses and execution paths.	Backend-performance comparison or universal portability.
RQ4	4 families; 240 sessions per policy; MCAD + 3 internal reference policies + 3 targeted ablations	Component sensitivity of the manipulated SAT/Real/ C_{eval} links under the controlled reference specification.	Not universal component necessity; no separately identified aggregate effect magnitude for no-Real versus relaxed C_{eval} ; no experimental superiority over literature methods.
RQ5	Structural factors 1, 5, 10, 20, 40	Structural growth of the CKG up to 1693 nodes and 2406 edges.	Throughput, concurrency, or industrial scalability of a distributed service.
RQ6	118000 observations; constraints, virtual nodes, membership density	Characterization of MCAD cost over the controlled ranges.	SQL/XMLA latency, multi-user contention, or end-to-end performance.

Table VI
DIAGNOSTIC READING OF THE CANONICAL Q1–Q6 TRACE FROM ALREADY ACQUIRED OBSERVATIONS. THE TABLE EXPLICITLY SEPARATES ADMISSIBILITY, MARGINAL COMPUTABILITY, AND PHYSICAL EFFECT; NO NEW EXECUTION IS INTRODUCED.

Step	SAT	Newly computable con- straint	$\Delta\varphi_t$	Action / physical	Reason	Scientific reading
Q1	true	c_1 (sales)	1/3	ALLOW / executed	new support	first newly computable constraint
Q2	true	c_2 (cost)	1/3	ALLOW / executed	new support	second newly computable constraint
Q3	true	c_3 (margin)	1/3	ALLOW / executed	new support	objective fully computable
Q4	true	–	0	BLOCK / not exe- cuted	outside objective scope	admissible, but no newly computable constraint
Q5	false	–	0	BLOCK / not exe- cuted	incompatible grain	inadmissible: the grain clause fails
Q6	true	–	0	BLOCK / not exe- cuted	redundancy after coverage	admissible and redundant: $\Delta\varphi = 0$

Table VII
PRIMARY-REASON PROFILE OVER THE 7960 FOODMART DECISIONS, REAGGREGATED FROM THE EXISTING CAMPAIGN.

Reason	n	% total	% BLOCK
ALLOW: new support	2266	28.5	–
BLOCK: primary redundancy reason $\Delta\varphi = 0$	3372	42.4	59.2
BLOCK: incompatible grain	770	9.7	13.5
BLOCK: empty subspace	576	7.2	10.1
BLOCK: incompatible slicer	566	7.1	9.9
BLOCK: outside objective scope	300	3.8	5.3
BLOCK: non-targeted measure	110	1.4	1.9

with three internal reference policies—a permissive naive policy, a measure-overlap heuristic, and a matched random policy—and three targeted ablations that each remove or relax one reasoning link. The reference policies are diagnostic comparators, not reimplementations of neighboring published systems. In this benchmark, *false ALLOW* means that the tested policy authorizes a query that the scenario specification marks for blocking; *false BLOCK* is the reverse. These are therefore deviations from a controlled operational reference, not errors against an independently observed notion of human utility or decision quality.

Within this controlled benchmark, MCAD preserves the final coverage of deterministic permissive policies while recording zero false ALLOW, false BLOCK, and non-contributive execution rates in every scenario family. Removing SAT raises mean false-ALLOW to 0.0417 and non-contributive execution to 0.125; removing Real or relaxing C_{eval} raises them to 0.08125

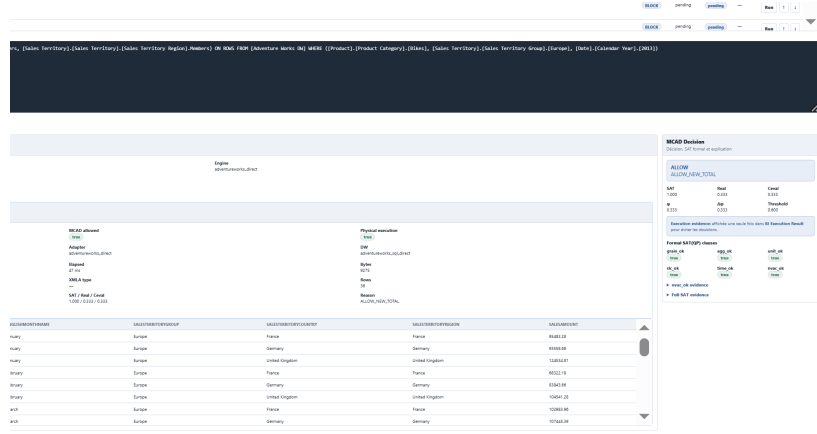
and 0.1875. The three ablations are interpreted component-wise: each manipulation degrades reference-aligned behavior in at least one exercised family, providing sensitivity evidence that the manipulated link is behaviorally consequential under this specification. This does not establish that every MCAD component is necessary in every setting. Moreover, the no-Real and relaxed- C_{eval} variants have identical aggregate false-ALLOW and non-contributive-execution rates here, so this protocol does not separately identify their effect magnitudes. None of these results constitutes direct superiority over neighboring systems.

E. Structural Growth and Semantic-Decision Cost

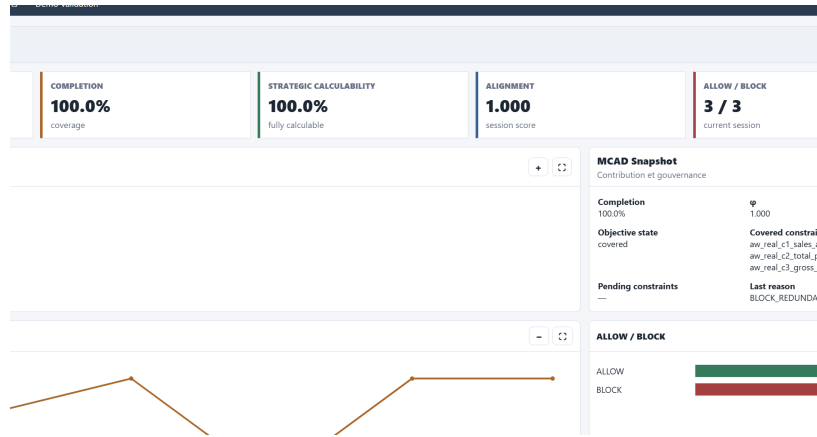
As the scale factor grows from 1 to 40, the CKG reaches 1693 nodes, 2406 edges, and 322 virtual nodes at the largest studied level. This experiment describes how the semantic representation grows. It is not a throughput test and should not be read as an industrial scalability result.

MCAD timing cost is characterized separately along three complexity dimensions with canonical measurements: number of constraints, number of virtual nodes, and membership density. p50 represents typical behavior, while p95 and p99 expose the tail of the distribution, which matters for an interactive layer [38]; the protocol also retains observation counts to avoid interpreting an isolated percentile [39].

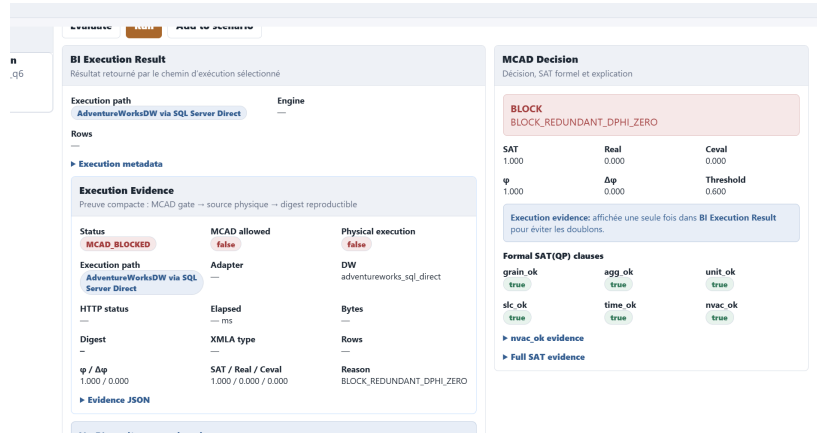
Virtual-node count shows the clearest timing trend, membership density only a weak effect, and constraint count is non-monotonic. From 6 to 24 virtual nodes, p50 rises from 0.3366 to 0.5223 ms (+55.2%) and p99 from 0.6488 to 1.1464 ms (+76.7%); from 25% to 100% membership density, p50 changes



(a) Q1: positive contribution, ALLOW, and physical execution



(b) After Q6: 3 ALLOW / 3 BLOCK and 100% objective computability



(c) Q6: admissible, redundant, $\Delta\varphi = 0$, without physical execution

Figure 5. End-to-end visual evidence under the strict demonstration policy. (a) Q1 brings a new support, receives ALLOW, and reaches the backend. (b) After the six queries, the session overview shows 3 ALLOW, 3 BLOCK, and 100% strategic computability. (c) Q6 remains contextually admissible but adds no new computability; the strict policy chooses BLOCK before the backend. All three panels are native captures from the same instrumented session.

from 0.4561 to 0.4652 ms (+2.0%) and p99 from 1.0850 to 1.1378 ms (+4.9%). These are descriptive transformations, not inferential effects. Across all published factors, p50 remains 0.337–0.666 ms and p99 0.649–1.461 ms; backend interaction, contention, dynamic CKG maintenance, and industrial-scale portfolios remain outside scope.

F. Results Synthesis

No single experiment is meant to carry the entire argument. Instead, the evidence is deliberately complementary. The Q1–Q6 trace exposes the session-level causal mechanism; the 1000-session FoodMart workload checks repeated enforcement of the strict policy before the backend; the multi-warehouse and

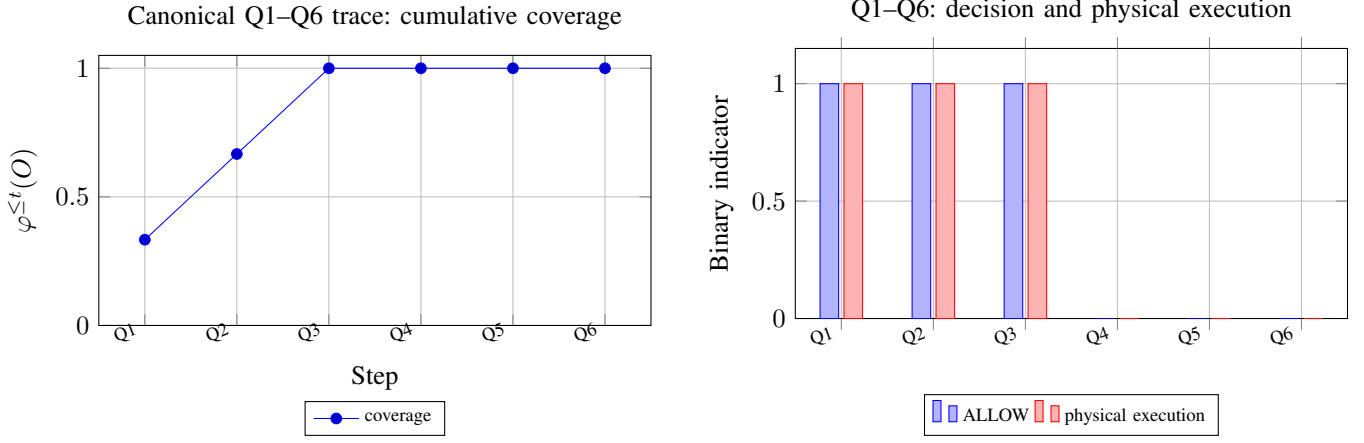


Figure 6. Q1-Q6 trace: (left) cumulative computability progress; (right) strict ALLOW/BLOCK policy instantiation of the indicator and corresponding physical effect.

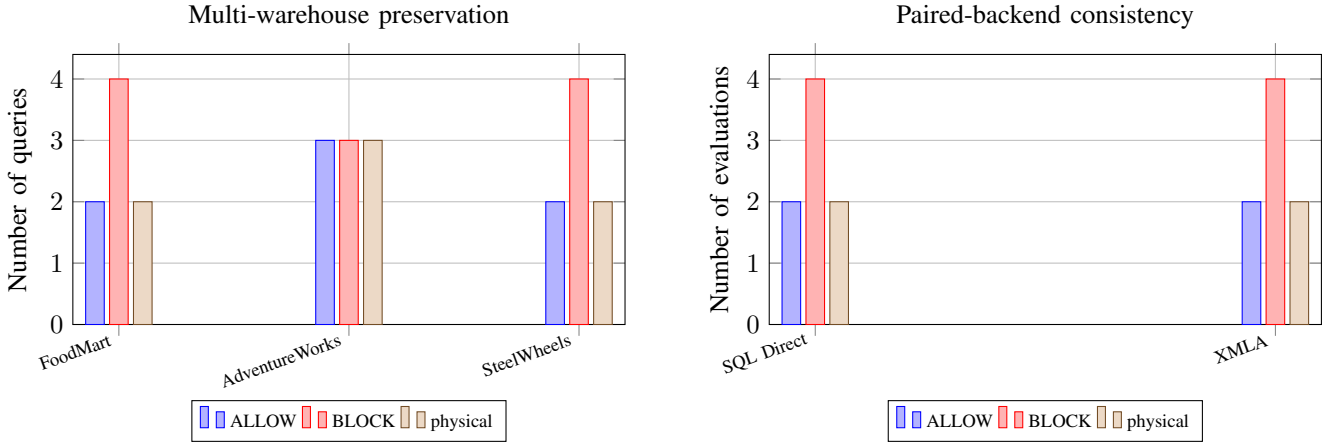


Figure 7. Decision-behavior preservation in the tested configurations: (left) FoodMart, AdventureWorks, and SteelWheels; (right) consistency of AdventureWorks decisions between SQL Direct and XMLA.

Table VIII
CONTROLLED ROBUSTNESS: COMPARISON WITH REFERENCE POLICIES AND ABLATIONS.

Policy	Sessions	Final φ	False ALLOW	False BLOCK	Non-contributive exec.
MCAD	240	0.500000	0.000000	0.000000	0.000000
Naive	240	0.500000	0.752083	0.000000	0.752083
Measure overlap	240	0.500000	0.660417	0.000000	0.731250
Matched random	240	0.119444	0.211597	0.185833	0.650556
Without SAT	240	0.500000	0.041667	0.000000	0.125000
Without Real	240	0.500000	0.081250	0.000000	0.187500
Relaxed C_{eval}	240	0.500000	0.081250	0.000000	0.187500

Table IX
STRUCTURAL GROWTH OF THE CKG.

Factor	Objectives	Constraints	Virtual nodes	Nodes	Edges	Snapshot (bytes)
1	3	8	10	55	66	43501
5	11	32	42	223	306	207117
10	21	62	82	433	606	412597
20	41	122	162	853	1206	822597
40	81	242	322	1693	2406	1642597

paired-backend studies examine preservation across different execution paths; the robustness benchmark checks component

sensitivity through internal reference policies and targeted ablations; and the structural and timing studies characterize

Table X
MCAD SEMANTIC-DECISION LATENCY QUANTILES.

Factor	Level	n	p50 (ms)	p95 (ms)	p99 (ms)	p99/p50
Constraint count	2	4000	0.6660	1.2822	1.4606	2.193
	4	4000	0.5081	0.9555	1.2348	2.430
	8	4000	0.4464	0.8285	0.9108	2.040
	12	4000	0.4650	0.8795	1.0596	2.279
	16	4000	0.4689	0.8837	1.0428	2.224
Virtual nodes	6	2000	0.3366	0.5929	0.6488	1.928
	12	2000	0.3917	0.7084	0.8679	2.216
	24	2000	0.5223	0.9624	1.1464	2.195
Membership density	25%	23000	0.4561	0.8975	1.0850	2.379
	50%	23000	0.4595	0.9074	1.1040	2.403
	75%	23000	0.4628	0.9108	1.1155	2.410
	100%	23000	0.4652	0.9169	1.1378	2.446

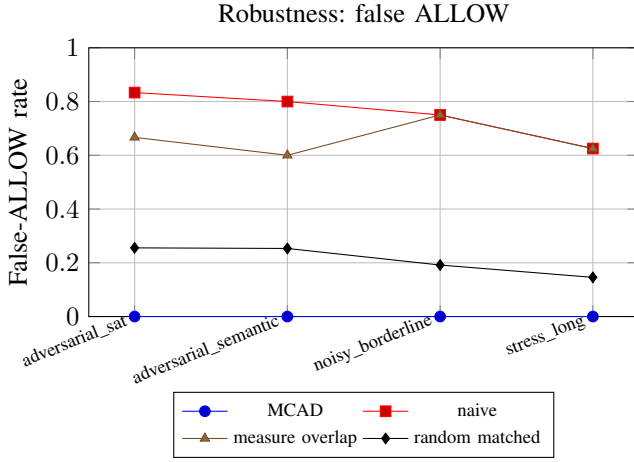


Figure 8. False-ALLOW rate by scenario family for MCAD and the internal reference policies.

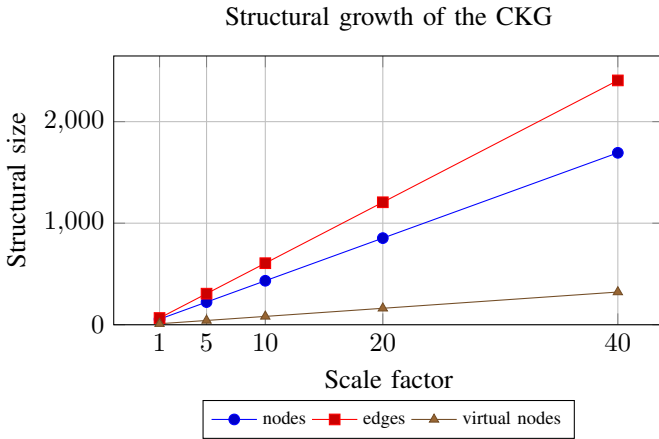


Figure 9. Structural growth of the CKG as a function of scale factor.

the cost of the semantic layer. Across these experiments, the important case $SAT = \text{true}$ with $\Delta\varphi = 0$ is observed and kept distinct from inadmissibility. Reaggregating the 7960 recorded decisions supports the same interpretation: a substantial share of BLOCK decisions carries the primary session-redundancy reason, while MCAD retains zero false ALLOW, false BLOCK,

and non-contributive execution in each robustness family relative to the benchmark specification. The empirical claim is therefore about computability, session dependence, and consistency of the mechanism; the operational response remains configurable.

This diversity reduces the risk that a result is specific to a single trace or backend without establishing universality. Interpretations are bounded to the objectives, warehouses, policies, and complexity factors explicitly covered by Table V; generalization to organizations, concurrent workloads, or dynamic objectives remains future validation work.

Evidence traceability: For each RQ, canonical data, derived artifacts, and the interpretation boundary are associated in the reproducibility package. Quantitative readings added through secondary analysis (FoodMart reason profile, Q1–Q6 diagnostic, scenario-family robustness view, and descriptive quantile variations) are computed only from already acquired observations and add no session, query, or timing measurement to the protocol.

VII. DISCUSSION, LIMITATIONS, AND FUTURE DIRECTIONS

The results support a central interpretation: query contribution in a decision-analysis session is not reducible to validity, preference similarity, or cognitive interest. MCAD instead measures variation in strategic-objective computability relative to context and session history, explaining why equally admissible queries can contribute differently and why repeated evidence can lose marginal value.

The CKG plays a stronger role than a metadata catalog. It brings together the decision-analysis context required for reasoning: multidimensional structures, business semantics, KPIs, constraints, sufficient supports, rules, temporal information, and cumulative state. Separating this context, the contribution indicator, and policy π_{act} allows the model to support several forms of assistance without changing its core semantics.

The contextual mask is not a filtering policy. It represents the semantic relation among the query plan, context, and objective supports; contribution is evaluated separately, after which an action policy may exploit that evaluation.

A. From Contribution Indicator to Interaction Strategies

BLOCK is not the default meaning of non-contribution but one strict governance choice. The same $\Delta\varphi = 0$ can

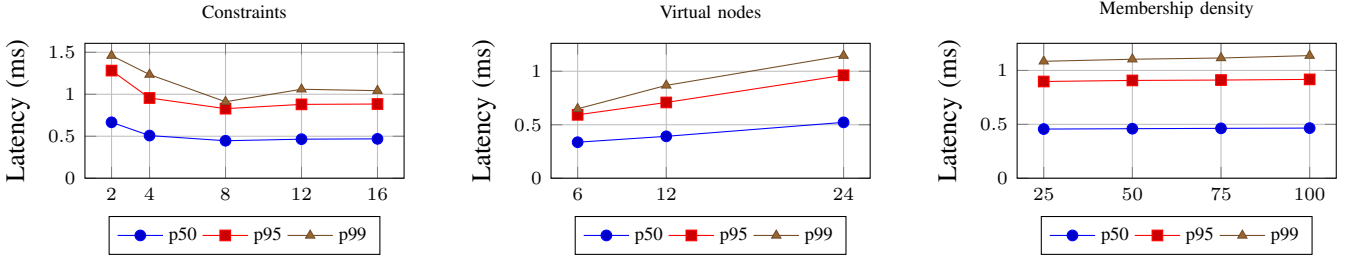


Figure 10. p50/p95/p99 semantic-decision latency as the number of constraints, virtual-node count, and membership density vary. Backend time is excluded.

be displayed, turned into a warning, used for guidance or recommendation, or blocked under compliance or cost-control policy. MCAD specifies *what the query contributes*; deployment policy decides *what to do with that information*.

B. Value Purpose and Complementarity with User-Centered Approaches

MCAD does not replace user-centered criteria. A query may be preferred, novel, or useful despite $\Delta\varphi_t = 0$, while an unsurprising query may be essential if it completes a strategic support. Computability and user-centered value are therefore complementary dimensions; the present experiments do not measure satisfaction, learning, or human decision quality.

Within a fixed semantic phase, objective, supports, and contextual rules remain fixed and acquired evidence E_t is monotone, so simple repetition does not restore contribution. The general session semantics permits several queries to jointly complete a support, although reported campaigns use single-query completion. Contribution may change in another session or after an explicit objective, scope, or rule change that starts a new phase.

C. Implications

From these results, three implications can be stated without extending the claims beyond the evaluated scope. First, MCAD provides a mechanism for **analytical-governance support** based on computability state rather than execution alone. Second, $\varphi(QP, O)$ and $\Delta\varphi_t$ can serve as explainable strategic-computability indicators complementary to personalization, guidance, or recommendation; they do not replace them. Third, explicit representation of context, sufficient supports, and marginal gains offers an interface to sequential and contextual-optimization policies [29], where strategic progress could become an interpretable reward or constraint.

D. Limitations and Threats to Validity

Construct validity and business oracle. MCAD measures *strategic computability*: the ability of a session to possess sufficient supports for evaluating objective constraints. This quantity is neither the final economic value of a decision nor analyst satisfaction. A query can make a KPI computable without changing a manager’s decision; conversely, qualitatively important information can remain outside the model if the objective is poorly operationalized. Objective decomposition and sufficient-support specification therefore form a semantic

oracle supplied to the mechanism; errors, ambiguity, or omissions can propagate to SAT, Real, C_{eval} , κ_c , and contribution measures. The campaigns verify the prototype relative to the supplied CKG, business rules, constraints, and supports; they do not independently validate the business correctness of that specification. Summarizability and provenance strengthen computation correctness and result traceability, respectively, without removing this dependence. Three levels must therefore be distinguished: reasoner correctness relative to encoded rules, correctness of those rules relative to business needs, and human utility of the resulting action. The κ_c/ψ intra-constraint extension is formalized but not experimentally evaluated in this study; the reported campaigns use $\varphi/\Delta\varphi$. Mask construction also depends on the quality of extraction to the canonical plan, sufficient-support specification, and relations encoded in the CKG. The present study validates reasoning over representations prepared for controlled scenarios; it does not yet validate automatic mask construction from arbitrary queries, business documents, or heterogeneous sources.

Experimental validity, external comparison, and performance. Physical evidence, the controlled robustness benchmark, and structural/temporal characterizations answer different questions. The robustness reference policies are diagnostic, and the ablations provide component-sensitivity evidence only for the manipulated SAT, Real, and C_{eval} links under the controlled specification; they do not establish universal component necessity. Djedaini, Negre, IAM, and other close neighbors were not reimplemented as competing systems, so no direct experimental superiority over them is claimed. Physical verification covers FoodMart, AdventureWorks, and Steel-Wheels, two backend paths (SQL Direct and XMLA/Mondrian), an instrumented trace, 1000 sessions, and controlled multi-configuration scenarios: it establishes multi-context feasibility and observed preservation within tested paths, not universal invariance. The p50/p95/p99 values cover 118000 semantic-decision observations and exclude SQL/XMLA latency, contention, and multi-user throughput; structural characterization describes CKG growth to scale factor 40 and is not an industrial capacity benchmark.

Action policy and human factors. ALLOW/BLOCK is a strict instantiation selected to verify that an MCAD-derived decision can act before the backend; it defines neither contribution nor the general framework policy. Notification, warning, guidance, recommendation, or permissive execution can use the same indicator. The current protocol does not measure acceptability, satisfaction, learning, or final human

decision quality; $\Delta\varphi$ is therefore not presented as a validated predictor of human utility.

E. Future Directions

Priority extensions are broader and more independent tests of the same model: automated CKG construction and maintenance using knowledge-graph techniques [20]; independent expert specification of objectives, constraints, and supports; support-level provenance summaries [35]; and multiple or conflicting objectives. Interaction studies could compare policies based on $\Delta\varphi/\Delta\psi$, while direct common-task comparisons, concurrent deployments, and user studies would address the main empirical questions left open.

VIII. CONCLUSION

MCAD introduces a contextual mask obtained by semantically mapping a candidate query to the decision-analysis context. In practical terms, the objective is made explicit through constraints and sufficient supports, the query is reduced to a canonical plan, and the mapping is then evaluated relative to what the session has already acquired. This makes it possible to state, explicitly and inspectably, whether the current query makes any objective constraint newly computable, rather than merely whether the query is executable or semantically admissible. Marginal contribution $\Delta\varphi$ keeps admissibility, evidence realization, and progress toward the objective separate, while the action itself remains a policy decision. The κ/ψ extension further captures partial intra-constraint computability, but that graded extension is formal rather than experimentally validated in this study.

Within the tested configurations, the prototype behaves consistently with the MCAD reasoning model, preserves decision behavior across exercised backend paths, and exhibits the measured semantic-decision cost. Controlled robustness and targeted ablations show that the manipulated reasoning links are behaviorally consequential under the benchmark reference specification; ALLOW/BLOCK remains only one policy instantiation.

These findings remain conditional on the supplied CKG, business rules, objectives, constraints, and supports. Independent business validity, human query value, interaction-policy effects, direct superiority over neighboring methods, and industrial-workload generalization are not established. MCAD should therefore be read as a formal, explainable layer for strategic computability, not as a validated measure of organizational decision quality.

APPENDIX A COMPLEMENTARY FORMAL PROOFS

Proof of Proposition 1: By definition $C_{\text{eval}}(QP, O) \subseteq C^*(O)$ and $|C^*(O)| > 0$. Hence $0 \leq |C_{\text{eval}}(QP, O)| \leq |C^*(O)|$, so $\varphi(QP, O) \in [0, 1]$. The two equality characterizations follow directly from finite-set cardinality and the inclusion $C_{\text{eval}}(QP, O) \subseteq C^*(O)$. ■

Proof of Proposition 2: Assume $\text{Real}(QP_a) \subseteq \text{Real}(QP_b)$. If $c \in C_{\text{eval}}(QP_a, O)$, some $R \in \mathcal{R}(c)$ satisfies $R \subseteq \text{Real}(QP_a)$, hence $R \subseteq \text{Real}(QP_b)$ and $c \in C_{\text{eval}}(QP_b, O)$. Therefore $C_{\text{eval}}(QP_a, O) \subseteq C_{\text{eval}}(QP_b, O)$ and $\varphi(QP_a, O) \leq \varphi(QP_b, O)$. ■

Proof of Proposition 3: Inside a fixed semantic phase, $E_{t-1} \subseteq E_t$. If $c \in C_E^{\leq t-1}(O)$, a sufficient support $R \subseteq E_{t-1}$ exists, hence $R \subseteq E_t$ and $c \in C_E^{\leq t}(O)$. Thus $C_E^{\leq t-1}(O) \subseteq C_E^{\leq t}(O)$. Dividing cardinalities by the positive constant $|C^*(O)|$ proves monotonicity and boundedness of $\varphi^{\leq t}$ and therefore $\Delta\varphi_t \geq 0$. ■

Proof of Proposition 4: For finite nonempty R , $0 \leq |R \cap X|/|R| \leq 1$. Taking a maximum over the finite nonempty family $\mathcal{R}(c)$ gives $\kappa_c(X) \in [0, 1]$, and averaging over finite nonempty $C^*(O)$ gives $\psi \in [0, 1]$. If $c \in C_{\text{eval}}(QP, O)$, at least one support is fully contained in $\text{Real}(QP)$ and therefore $\kappa_c = 1$; averaging yields $\varphi(QP, O) \leq \psi(QP, O)$. If $E_{t-1} \subseteq E_t$, support intersections can only grow, hence each $\kappa_c(E_t)$ and their average are non-decreasing, so $\Delta\psi_t \geq 0$. ■

Proof of Proposition 5: Assume (A1) and (A2). Because evidence is retained monotonically within the phase, $C_E^{\leq i}(O) \subseteq C_E^{\leq t}(O)$ for $i \leq t$. The initial term $C_E^{\leq 0}(O)$ is therefore contained in $C_E^{\leq t}(O)$. Moreover, by (A1), for each $i \leq t$, $C_{\text{eval}}(QP_i, O) \subseteq C_E^{\leq i}(O) \subseteq C_E^{\leq t}(O)$. Taking the union gives $C_Q^{\leq t}(O; E_0) \subseteq C_E^{\leq t}(O)$.

For the reverse inclusion, let $c \in C_E^{\leq t}(O)$. If $c \in C_E^{\leq 0}(O)$, then $c \in C_Q^{\leq t}(O; E_0)$ by definition. Otherwise there is a least $j \in \{1, \dots, t\}$ such that $c \in C_E^{\leq j}(O)$, hence $c \in C_E^{\leq j}(O) \setminus C_E^{\leq j-1}(O)$. By (A2), $c \in C_{\text{eval}}(QP_j, O)$, so $c \in C_Q^{\leq t}(O; E_0)$. Equality follows.

If only (A1) is retained, the first inclusion remains valid. It can be strict: let one constraint have support $R = \{nv_1, nv_2\}$, $E_0 = \emptyset$, $A_1 = \{nv_1\} \subseteq \text{Real}(QP_1)$, and $A_2 = \{nv_2\} \subseteq \text{Real}(QP_2)$. Then $R \subseteq E_2$ but no individual query realizes R , so the constraint belongs to $C_E^{\leq 2}(O)$ and not to $C_Q^{\leq 2}(O; E_0)$. ■

APPENDIX B DETAILED FORMALIZATION OF THE FOODMART HF/NORTH EXAMPLE

We return to the FoodMart example used to illustrate the mechanism's formalization. The *Sales* cube is defined by:

- $F = \{\text{Sales}\}$,
- three dimensions $D = \{\text{Time}, \text{Product}, \text{Store}\}$, and
- measures $M = \{\text{Store Sales}, \text{Store Cost}, \text{Units Sold}\}$ together with a derived KPI Margin%.

The dimension hierarchies are:

- Time: Day $\rightarrow$ Month $\rightarrow$ Year,
- Product: Product $\rightarrow$ Subcategory $\rightarrow$ Category,
- Store: Store $\rightarrow$ City $\rightarrow$ Region.

The objective $O_{\text{HF_NORTH}}$ is defined by the constraints formalized below, with $C^*(O_{\text{HF_NORTH}}) = \{c_1, c_2, c_3\}$. Each constraint is associated with a set of virtual nodes:

- $N(c_1)$ contains nodes of the form

$$nv_{1,m} = (\text{Sales}, g_m, \text{Margin}\%, \text{avg}, \%, S_{\text{HF_North}}, w_{1998}).$$

where g_m has grain {Month, Category=Health Food, Region=North}, $S_{HF,North}$ restricts the category to Health Food and the region to North, and w_{1998} is the temporal window corresponding to year 1998.

- $N(c_2)$ contains nodes of the form

$$nv_{2,r} = (Sales, g_r, StockoutRate, ratio, \%, S_{HF,North}, w_{1998}).$$

where g_r has grain {Store, Category=Health Food, Region=North} and StockoutRate is a KPI derived from *Units Sold* and inventory information.

- $N(c_3)$ contains nodes of the form

$$nv_{3,m} = (Sales, g_m, Sales, sum, USD, S_{HF,North}, w_{1997-1998}).$$

where Sales = Store Sales and $w_{1997-1998}$ covers years 1997 and 1998.

Illustration of partial intra-constraint computability:

This extension is purely formal in the present version and does not reinterpret any experimental campaign. To illustrate κ_c , suppose that the minimal support of c_1 requires one monthly cell for each of the twelve months of 1998. If the current state E_t contains six of these twelve nodes and no other more advanced sufficient support for c_1 , then

$$\kappa_{c_1}(E_t) = \frac{6}{12} = 0.5.$$

Constraint c_1 is not yet fully computable, but its internal progress is no longer zero. Once all twelve required nodes have been acquired, $\kappa_{c_1} = 1$ and the support of c_1 is complete under the graded extension. This example therefore explicitly distinguishes *partial coverage of the objective from partial computability within a single constraint*, without modifying the binary values reported in the existing campaigns.

Consider, for example, the guided query QP'_1 :

$$QP'_1 = (Sales, g_m, \{Margin\%\}, \{avg\}, \{\%\}, S_{HF,North}, w_{1998}).$$

Under reasonable CKG compatibility assumptions,

$$\text{Real}(QP'_1) \supseteq \{nv_{1,m} \mid m \in \text{months of 1998}\},$$

and therefore $c_1 \in C_{\text{eval}}(QP'_1, O_{HF_NORTH})$. If no cell from $N(c_2)$ or $N(c_3)$ is realizable by QP'_1 , then

$$C_{\text{eval}}(QP'_1, O_{HF_NORTH}) = \{c_1\},$$

$$\varphi(QP'_1, O_{HF_NORTH}) = \frac{1}{3}.$$

Likewise, queries QP'_2 and QP'_3 are constructed to realize nodes from $N(c_2)$ and $N(c_3)$, respectively, allowing the session (QP'_1, QP'_2, QP'_3) to reach complete coverage of the objective constraints quickly.

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